

Code:

```
#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;
int main() {
    int n, frames;
    cout << "Enter number of page references: ";
    cin >> n;
    vector<int> pages(n);
    cout << "Enter page reference string:\n";
    for (int i = 0; i < n; i++)
        cin >> pages[i];
    cout << "Enter number of frames: ";
    cin >> frames;
    vector<int> memory;
    vector<int> recent;
    int pageFaults = 0, pageHits = 0;
    cout << "\nPage\tFrames\n";
    for (int i = 0; i < n; i++) {
        int page = pages[i];
        auto it = find(memory.begin(), memory.end(), page);
        if (it != memory.end()) {
            pageHits++;
            recent.erase(find(recent.begin(), recent.end(), page));
            recent.push_back(page);
        }
        else
        {
            pageFaults++;
            if (memory.size() < frames) {
                memory.push_back(page);
            }
            else
            {
                int lru = recent.front();
                recent.erase(recent.begin());
                auto pos = find(memory.begin(), memory.end(), lru);
                *pos = page;
            }
            recent.push_back(page);
        }
        cout << page << "\t";
        for (int x : memory)
            cout << x << " ";
    }
```

```

cout << endl;
}
cout << "\nTotal Page Hits = " << pageHits << endl;
cout << "Total Page Faults = " << pageFaults << endl;
cout << "Page Fault Ratio = " << (float)pageFaults / n << endl;

return 0;
}

```

Output:

```

Enter number of page references: 12
Enter page reference string:
7 0 1 2 0 3 0 4 2 3 0 3
Enter number of frames: 3

Page    Frames
7        7
0        7 0
1        7 0 1
2        2 0 1
0        2 0 1
3        2 0 3
0        2 0 3
4        4 0 3
2        4 0 2
3        4 3 2
0        0 3 2
3        0 3 2

Total Page Hits = 3
Total Page Faults = 9
Page Fault Ratio = 0.75

```